Colorimetric measurement of œstrogen in male and post-menopausal urine is frequently near the limits of sensitivity of the method. Concentration of feeble colours by extraction into chloroform–trichloroacetic acid should enable low levels of œstrogen to be measured spectrophotometrically with greater confidence, and the spectral shift is a check on the authenticity of the Kober colour. It is suggested that the following type of procedure may be valuable as a supplement to the conventional analytical procedures for œstrogen utilizing the Kober reaction[3,4]. For œstrone or its methyl ether: dilute a 3-ml. aliquot of the Kober colour (after reading) with 1·8 ml. of 2 *N* trichloroacetic acid in a pointed centrifuge tube, stopper, mix, cool under the tap, add 1·2 ml. of chloroform, shake for 20 sec, centrifuge briefly, suck off top layer and discard, read the bottom layer in a 1 ml. cuvette at 502, 532, 562 m*M* (508, 538, 568 m*M* for the ether) in a spectrophotometer. This gives an Allen corrected value about three times as great as that derived from measurements on the original Kober colour. The chloroform extract fades about 3 per cent in ½ hr. Tetrachloroethane is objectionably toxic, though an excellent solvent.

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Controlled Use of Adsorbents in Enzyme Purification

Published methods for the purification of many enzymes specify the use of $C\gamma$-aluminium hydroxide gel[1] or calcium phosphate gel[2] as adsorbents, and Colowick[3] has reviewed the use of these methods.

The adsorbents may be used (*a*) preferentially to adsorb contaminants from solution or (*b*) preferentially to adsorb the enzyme from solution. These may be regarded as the two gross extremes of (*c*) a graded fractional adsorption, and Ochoa[4] has used all three procedures for the purification of the 'malic' enzyme from different sources.

However, the use of fractional procedures is reported only rarely in the literature, and most authors seem to make use of a technique which superficially resembles (*b*). This procedure is carried out in such a way as to afford little chance for investigating, and thus utilizing, the possibly fine resolving power of the adsorbents. The gel suspension is usually added in one or two large batches to adsorb all the enzyme and, after sedimentation, the gel is washed and eluted.

In this communication the behaviour is reported of two different enzyme preparations on different gels, and evidence is presented of differential adsorption of protein during the procedure. The results emphasize the importance of fractional adsorption in enzyme purification with gels, and indicate that this technique can lead to greater purification than is possible using the popular technique above.

The method requires titration of the enzyme remaining in solution after addition of small batches of adsorbent, and a rapid method for the assay of enzymatic activity is desirable for convenient working.

(1) *Preparation of glucose-6-phosphate dehydrogenase.* This enzyme can be conveniently prepared from dried baker's yeast, but is usually contaminated with hexokinase. In a method used in this laboratory (unpublished), the hexokinase activity is destroyed by mild-heat treatment of a yeast autolysate and nucleic acids are precipitated by protamine sulphate. Some impurities are removed by iso-electric precipitation and the crude enzyme is fractionated with ammonium sulphate. Treatment with $C\gamma$ alumina results in further purification.

Table 1 shows the results obtained in this step by the popular procedure and the fractional method.

Table 1. Fractionation of Glucose-6-Phosphate Dehydrogenase with Alumina Gel

Method 1:
Enzyme adsorbed from 0·05 *M* acetate buffer *p*H 5·5 with 2·5 ml. of $C\gamma$ alumina (11 mgm. solids/ml.). Gel precipitate collected by centrifugation, washed with 0·05 *M* acetate *p*H 5·5, eluted three times with 0·1 *M* phosphate buffer *p*H 7·4.

Volume of gel added (ml.)	Total units* of enzyme in solution	Total protein† (mgm.)	Specific activity (units/mgm. protein)
0	260	20	13
2·5	31	—	—
Alumina eluate	230	9·7	24

Method 2:
Enzyme solution in 0·05 *M* acetate buffer *p*H 5·5, 29·0 ml. of $C\gamma$ alumina suspension added. Gel precipitate discarded, 30 ml. of $C\gamma$ alumina added to enzyme solution. Gel collected by centrifugation washed and eluted as above.

Volume of gel added (ml.)	Total units* of enzyme in solution	Total protein† (mgm.)	Specific activity (units/mgm. protein)
0	2,430	200	12
29	2,300	—	—
30	36	—	—
Alumina eluate	2,010	34	59

* Determined by the rate of reduction of triphosphopyridine nucleotide ($0·2 \times 10^{-3}$ *M*) in the presence of *tris* (hydroxymethyl)-aminomethane buffer (0·03 *M*, *p*H 7·4), magnesium chloride (3×10^{-3} *M*) and glucose-6-phosphate (10^{-3} *M*) in a Beckman *DU* spectrophotometer at 30° C. The unit of dehydrogenase activity is defined as an optical density change of 0·100 per min. at 340 mμ and 30° C.
† Determined by the biuret method (ref. 5).

These results indicate that the fractional procedure gives a superior purification factor in this case. In method 1, the specific activity of the enzyme increases by a factor of about 1·9, while in the second method the increase in the factor is about 5.

(2) *Preparation of uridinediphosphoglucose (UDPG) dehydrogenase.* This enzyme is prepared from fresh calf-liver by a modification of a method previously applied to acetone powders of liver[6].

Fresh calf-liver is treated in a Waring blender with 0·15 *M* potassium chloride and centrifuged. The supernatant solution is fractionated with ammonium sulphate, heat treated and fractionated again with ammonium sulphate. Calcium phosphate gel is then used in the fractional adsorption procedure.

In this step, the differential adsorption occurs of some other system (of unknown type) which interferes strongly with the assay method for UDPG dehydrogenase. This is demonstrated by the data in Table 2. As more gel suspension is added, the apparent activity of the dehydrogenase in solution rises to a peak value, and then begins to fall. At the peak value

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Table 2. Fractional Adsorption of UDPG-Dehydrogenase with Calcium Phosphate Gel

Adsorption of enzyme from solution in 0·01 *M* phosphate *p*H 6·8	
Total volume of gel added (ml.)	Total units* of enzyme in solution
0	12·4
9·0	36·3
18·0	49·5
30·0†	60·2
3·0	50·5
6·0	46·5
21·0	31·5
51·0	0

* Measured by the rate of reduction of diphosphopyridine nucleotide (7×10^{-4} *M*) in presence of glycine buffer (0·15 *M*, *p*H 8·7) and UDPG ($1{\cdot}5 \times 10^{-4}$ *M*). Micro-cells with total volumes of 0·7 ml. were used in a Beckman *DU* spectrophotometer at 30° C. The unit of enzyme activity is defined as a change in optical density at 340 mμ of 0·100 per min. at 30° C.

† At this point all the added gel was discarded ; 23 per cent of the total protein had disappeared from solution.

in a typical experiment, 23 per cent of the total protein had been adsorbed to the gel.

It can be seen that the fractional procedure brings about removal of an interfering system as previously mentioned, together with a substantial amount of protein. During the latter part of the step, the enzyme is adsorbed to the gel. The gel is washed with 0·01 *M* phosphate *p*H 6·8 and the enzyme eluted with 0·1 *M* phosphate buffer *p*H 7·4. Quantitative recovery of the enzymes (peak values) have been obtained in these experiments.

The results indicate that in procedures for the purification of other enzymes or proteins which involve the use of adsorption-elution techniques, the fractional methods described here may be of considerable value. Such procedures are relatively easy to apply provided that rapid assay methods are available.

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The Pasteur Effect in a Respiration-deficient Mutant of Baker's Yeast

The respiration-deficient mutant of baker's yeast, *Saccharomyces cerevisiae R XII A*, induced by high-temperature cultivation, was used for examining the point of action of oxygen in the Pasteur effect. The mutant was identical with the *petite colonie* mutant described by Ephrussi[1] and Raut[2]. The argument was that, if oxygen acted directly on some enzyme involved in glucose degradation (for example, glyceraldehyde phosphate dehydrogenase), it would decrease the rate of glucose catabolism even in the mutant lacking essential components of the cytochrome system.

It was found that in the presence of glucose there was no demonstrable uptake of oxygen by the suspension and that the Q_{CO_2} (=100–130) was not affected by the introduction of oxygen into the suspension. The uptake of glucose, as measured by

Table 1. Inorganic Phosphate and Labile Phosphate Levels in the Respiration-Deficient Mutant (μgm. Phosphorus/mgm. Dry Weight ± *S.E.*)

Incubation conditions	Inorganic phosphate	Labile phosphate
No substrate, anaerobically	4·24 ± 0·24	5·84 ± 0·31
No substrate, 10 min. aeration	3·63 ± 0·26	6·39 ± 0·67
5 per cent glucose, anaerobically	2·37 ± 0·23	7·15 ± 0·71
5 per cent glucose, 10 min. aeration	2·40 ± 0·21	7·26 ± 0·75

quantitative analysis as well as by distribution of radioactivity following application of universally labelled ^{14}C-glucose, revealed no differences between aerobic and anaerobic conditions. As another criterion of the Pasteur effect, the change of inorganic phosphate-level was examined. Here again no change due to oxygen could be observed to take place in the steady-state level in the presence of glucose (Table 1).

If these results can be generalized they may serve as a weighty support for the theory that oxygen brings about the Pasteur effect not by affecting directly any enzyme involved in the anaerobic breakdown of glucose but rather by making it possible for other catabolic pathways to function (much in line with Lynen's views[3,4]).

It was observed, however, that under conditions when no substrate was present in the medium, oxygen was effective in producing changes in the amounts or, at any rate, in the distribution of some metabolically active components of the mutant cell. Some of these changes (for example, in the penetration of free glucose into the cell) are dealt with elsewhere[5]. It was found here that oxygen decreased the level of inorganic phosphate in the absence of glucose, with simultaneous production of labile phosphate bonds (Table 1), the process being completely blocked by 10^{-4} *M* potassium cyanide but only partly inhibited (at most by 20 per cent of the steady-state level change) by 5×10^{-4} *M* 2,4-dinitrophenol. The generation of high-energy phosphate bonds was accompanied by only a minute oxygen consumption ($Q_{O_2} = 0{\cdot}25$–$0{\cdot}32$; $Q_{CO_2} = 1{\cdot}09$–$1{\cdot}48$) so that the apparent phosphorus : oxygen ratio reached values far exceeding those commonly reported for whole cells and even for purified enzyme systems. Thus endogenous phosphorylation in the mutant (and very likely in ordinary baker's yeast, ref. 6) appears to be qualitatively different from phosphorylation proceeding with glucose, and produces high-energy bonds at the expense of other energy sources (physicochemical state of proteins, nucleic acids ?) than those involved in oxidative phosphorylation known heretofore.

Detailed results will be published in *Folia microbiologica*.

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